

Quang-Vinh Dang
British University Vietnam
Hung Yen, Vietnam
vinh.dq4@buv.edu.vn

To the Editors,
Computational Statistics and Data Analysis

Dear Editors,

I am submitting the manuscript “WAGER: Within-cell Antisymmetric Gain Evaluation of Resolution” for consideration as a regular paper in *Computational Statistics and Data Analysis*.

The paper concerns the comparison of two probabilistic predictors when the labels are partly determined by a nuisance feature both models observe. In that situation the score difference confounds two quantities: a closer fit to the conditional label distribution of the nuisance feature, and genuinely better prediction of individual cases. Significance testing on the difference does not separate them, and the customary remedy — fitting a nuisance-only baseline and comparing against it — introduces estimation error, smoothing choices and model-family mismatch into what should be an exact statement about a fixed pair. Machine-learning benchmarks in computer vision supply the applications here, but the estimation problem is general.

WAGER answers that question directly, and the contribution is primarily methodological. Given two frozen probabilistic models, it forms their proper-score contrast, matches test examples sharing a declared prior feature, and crosses their labels. This yields an exact finite-sample identity — the observed gain equals a prior-transported gain plus an antisymmetric instance-alignment gain — whose residual is an order-two U-statistic that, under the quadratic score, equals the within-cell prediction–label covariance the new model has gained, linking it to the classical resolution term of proper-score decomposition. The estimator runs in $O(NK)$ time on cached predictions, fits no baseline model, has no tunable hyperparameter, and comes with cluster-robust intervals and a cell-stratified randomization test. Two further exact results are proved: leave-one-out transport removes an attenuation bias of $(n_c - 1)/n_c$ that a naive in-sample plug-in incurs, and coarsening the audit cell shifts the estimate by an explicit between-cell covariance term.

Because the method is a measuring instrument, the paper takes its validation seriously. Beyond recovery and null designs, we verify the coverage of the primary interval in a regime that stresses it, and we report discriminant-validity experiments establishing what the two channels do and do not separate — including an honest limitation, that a monotone recalibration moves score mass between the channels, together with the calibration-matched protocol that removes it and a demonstration that the protocol works.

The empirical results show the instrument earning its keep, and the clearest demonstration concerns a method the field already compares rather than any predictor of ours. We evaluate the released causal MOTIFS checkpoint of Tang et al. (CVPR 2020) in both of its modes with the authors’ own code on the standard Visual Genome PredCls split, reproducing their published behaviour, and decompose the debiased model against the biased baseline. Read naively, the Total Direct Effect variant appears to lose a little instance alignment. That reading does not survive the control this paper argues for: the two models differ sharply in confidence, and once temperatures are matched on held-out images the alignment channel is statistically indistinguishable from zero (-0.00006 , 95% CI $[-0.00120, +0.00109]$), with essentially the entire proper-score difference lying in the prior channel. A widely used debiasing method therefore buys ten points of mean recall with no measurable change in within-cell instance alignment

— which is what its construction subtracts, so the finding characterises the method rather than indicting it, while showing that a metric read as evidence about rare predicates can move sharply without any change in instance-level evidence.

Two further results point in complementary directions. A model reading real pixels through a frozen CLIP encoder scores 0.04655 *lower* overall — an evaluator reading that number would discard it — yet aligns with individual instances significantly *better* than the model that beats it, its entire deficit lying in the annotation prior and its advantage surviving both post-hoc prior correction and calibration matching. Applied unchanged to long-tailed image classification across seeds and imbalance ratios, a calibration-matched reanalysis overturns what the raw decomposition appears to say about two re-weighting schedules.

I believe the work belongs in *Computational Statistics and Data Analysis* under Section II, Statistical Methodology for Data Analysis, whose remit includes classification, pattern recognition, model selection and assessment, and cross-validation — the family this estimator joins. It also touches Section I through the development, evaluation and validation of an algorithm, which is supplied and documented rather than merely described. The contribution is a decomposition with exact finite-sample results and inference, validated as an instrument before being used as one; the vision benchmarks are the applications on which it is exercised, not the claim. No state-of-the-art result is asserted, and none is needed for the argument.

All code, cached model outputs and scripts needed to reproduce every number, table and figure are publicly available at <https://github.com/vinhqdang/WAGER>, including a verification script that checks each reported value against the committed results files.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. I am the sole author and approve its submission. I declare no competing financial or personal interests, and the work received no specific grant from any funding agency. In accordance with the journal’s policy, the manuscript includes a declaration of generative AI use in its preparation.

Thank you for your consideration.

Yours sincerely,

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam